

TASK 3: DATA VISUALIZATION USING PYTHON

INTRODUCTION

Data Visualization was performed using Python to represent data in the form of charts and graphs.

The main objective of this project is to convert raw data into meaningful visual representations that help in identifying trends, comparisons, and patterns easily.

Tools:

- Python
 - Visual Studio Code
 - Pandas
 - Matplotlib
 - Seaborn
 - Excel Dataset
-

DATASET DESCRIPTION

The dataset contains sales-related information used for analyzing product performance and business trends.

The dataset includes attributes such as:

- Product Name
- Category
- Sales
- Profit

- Region

The dataset was visualized using different graphical techniques to understand the data clearly.

STEPS PERFORMED

1. Imported required libraries

Python visualization libraries such as Matplotlib and Seaborn were imported along with Pandas for data handling.

2. Loaded dataset

The sales dataset was imported into Python for visualization and analysis.

3. Explored dataset information

The dataset was checked to understand the available columns and data structure.

4. Created sales visualization

A bar chart was created to compare sales performance across different categories.

5. Created profit analysis visualization

Graphs were generated to analyze profit contribution of different products.

6. Created region-wise analysis

Visualizations were created to compare sales distribution across different regions.

7. Created distribution plots

Histogram visualization was created to understand the distribution of sales values.

8. Analyzed patterns

The created charts helped identify trends, comparisons, and important insights from the dataset.

OBSERVATIONS

- Data visualization helped in understanding sales patterns easily.
 - Different categories showed different performance levels.
 - Product-wise profit comparison was successfully analyzed.
 - Graphical representation made data interpretation easier.
 - Visualizations helped identify important trends and relationships.
-

CONCLUSION

Data Visualization was successfully performed using Python.

The project helped convert raw data into meaningful graphical representations. The created charts and graphs provided better understanding of sales, profit, and category-wise performance.

Visualization techniques make complex datasets easier to analyze and support better decision-making in data analytics.

Author

Dasari Sanutha

CodeAlpha Data Analytics Internship